1. Display the absolute path of the current directory:

```
pwd
```

2. List all files, including hidden ones, with details and human-readable sizes:

```
ls -lah /home/ubuntu
```

3. Create the nested directories, including any missing parent directory:

```
mkdir -p ~/ds_lab/week1
```

4. Change to ~/ds_lab/week1:

```
cd ~/ds_lab/week1
```

5. Create an empty file:

```
touch experiment.log
```

6. Copy it to ~/ds_lab/, keeping the original:

```
cp experiment.log ~/ds_lab/
```

7. Rename it inside ~/ds_lab/week1:

```
mv ~/ds_lab/week1/experiment.log ~/ds_lab/week1/backup.log
```

8. Append the two lines to experiment.log:

```
printf 'line01\nline02\n' >> experiment.log
```

9. Display the entire file:

```
cat experiment.log
```

10. Display lines containing 02, with line numbers:

```
grep -n '02' experiment.log
```

11. Find regular files ending in .log under /home/ubuntu:

```
find /home/ubuntu -type f -name '*.log'
```

12. Set owner permissions to read/write, group permissions to read, and no permissions for others:

```
chmod 640 backup.log
```

13. Set the file's owner and group to root:

```
sudo chown root:root backup.log
```

14. Display all processes with detailed information, including process IDs and users:

```
ps aux
```

15. Display disk usage for all mounted filesystems in human-readable units:

```
df -h
```